Fixed-difference Kummer collector

KUMMER-DIFFERENTIAL-COLLECTOR-20261022

Completed targets per arm: 1260.

Uncached: 1260 source-verified, 230,430,039 quotient steps.

Uncached: 109,480,302 signed splits, 739 Recover failures.

Uncached cold ratio: 8.9597–55.9354; wins/losses: 0/36 cells.

Cold setup: 21 unique model builds, charged in all 1260 targets.

Source transfers and source verifications: 1260 each.

Every candidate and rho answer was independently checked as k·P=Q

on the original elliptic curve over Fp^2.

Rho ran in the same paired_uncached process.

These ratios use fresh paired rho counts.

Model setup includes root search, theta search and torsion matrices.

Each cold target includes source transfer, initialization, the

quotient walk, signed Recover, graph elimination and verification.

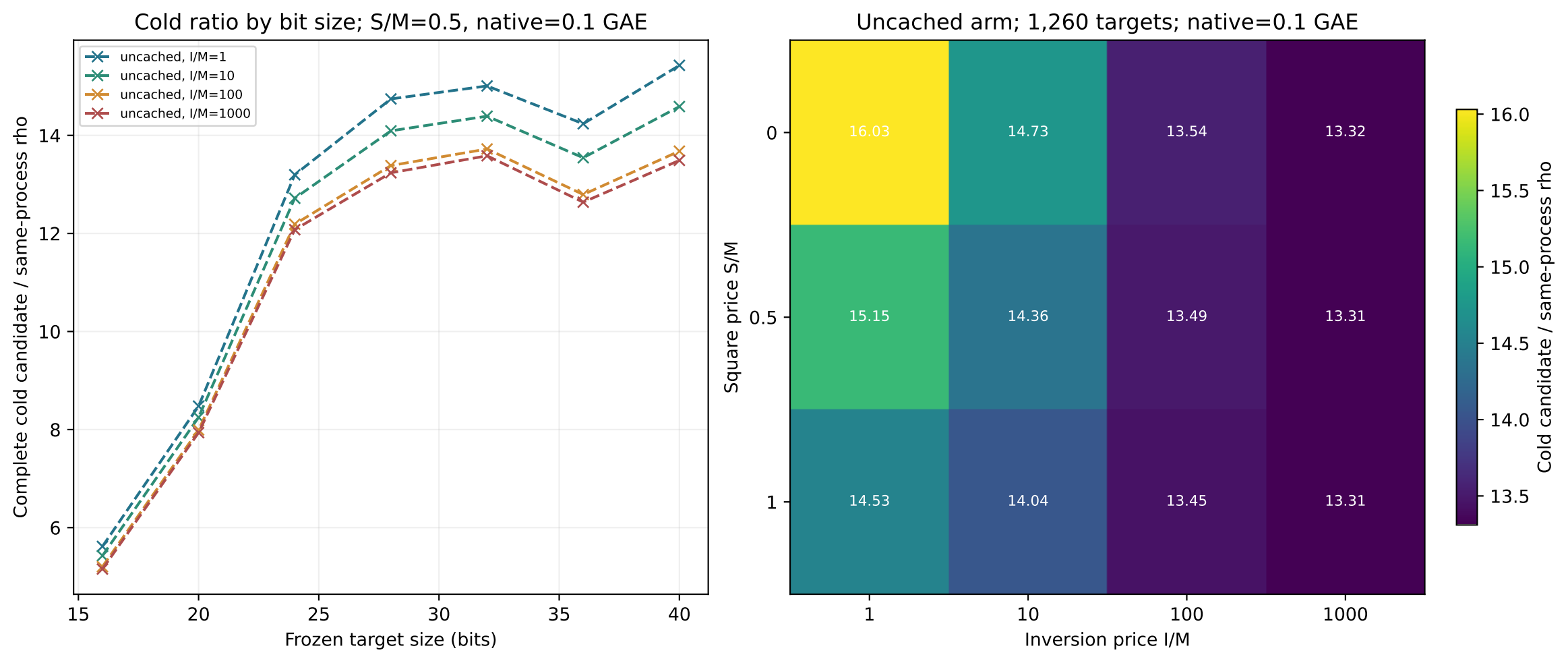
Raw: results/paired/<arm>/collector-*.jsonl, paired/rho/*.jsonl,

results/paired/analysis/{target-price,price-grid}.csv.

All 36 cells use S/M in {0,0.5,1}, I/M in {1,10,100,1000},

and native cost in {0,0.1,1} GAE per declared logical operation.

Ratios describe this measured implementation and frozen cohort.


Cold ratio by bit size; S/M=0.5, native=0.1 GAE
uncached, I/M=1
uncached, I/M=10
uncached, I/M=100
uncached, I/M=1000
Complete cold candidate / same-process rho
Frozen target size (bits)
Uncached arm; 1,260 targets; native=0.1 GAE
Square price S/M
Inversion price I/M
16.03
14.73
13.54
13.32
15.15
14.36
13.49
13.31
14.53
14.04
13.45
13.31
Cold candidate / same-process rho

uncached: native=0 GAE/logical

S/M	I/M	candidate GAE	rho GAE	ratio	wins/losses
0.0	1	9,798,151,036.8	175,168,978.8	55.935	0/1260
0.0	10	8,378,420,339.1	174,640,420.7	47.975	0/1260
0.0	100	7,075,915,111.8	174,155,505.1	40.630	0/1260
0.0	1000	6,830,646,535.8	174,064,192.9	39.242	0/1260
0.5	1	8,841,142,700.8	175,041,121.6	50.509	0/1260
0.5	10	7,966,770,880.2	174,617,698.3	45.624	0/1260
0.5	100	7,021,504,047.1	174,159,943.5	40.316	0/1260
0.5	1000	6,825,157,820.6	174,064,860.9	39.210	0/1260
1.0	1	8,157,565,318.0	174,949,795.0	46.628	0/1260
1.0	10	7,626,712,631.6	174,598,927.7	43.681	0/1260
1.0	100	6,969,019,037.8	174,164,224.7	40.014	0/1260
1.0	1000	6,819,690,778.5	174,065,526.2	39.179	0/1260

uncached: native=0.1 GAE/logical

S/M	I/M	candidate GAE	rho GAE	ratio	wins/losses
0.0	1	17,462,948,021.6	1,089,353,650.8	16.031	0/1260
0.0	10	16,043,217,323.9	1,088,825,092.7	14.734	0/1260
0.0	100	14,740,712,096.6	1,088,340,177.1	13.544	0/1260
0.0	1000	14,495,443,520.6	1,088,248,864.9	13.320	0/1260
0.5	1	16,505,939,685.6	1,089,225,793.6	15.154	0/1260
0.5	10	15,631,567,865.0	1,088,802,370.3	14.357	0/1260
0.5	100	14,686,301,031.9	1,088,344,615.5	13.494	0/1260
0.5	1000	14,489,954,805.4	1,088,249,532.9	13.315	0/1260
1.0	1	15,822,362,302.8	1,089,134,467.0	14.527	0/1260
1.0	10	15,291,509,616.4	1,088,783,599.7	14.045	0/1260
1.0	100	14,633,816,022.6	1,088,348,896.7	13.446	0/1260
1.0	1000	14,484,487,763.3	1,088,250,198.2	13.310	0/1260

uncached: native=1 GAE/logical

S/M	I/M	candidate GAE	rho GAE	ratio	wins/losses
0.0	1	86,446,120,884.8	9,317,015,698.8	9.278	0/1260
0.0	10	85,026,390,187.1	9,316,487,140.7	9.126	0/1260
0.0	100	83,723,884,959.8	9,316,002,225.1	8.987	0/1260
0.0	1000	83,478,616,383.8	9,315,910,912.9	8.961	0/1260
0.5	1	85,489,112,548.8	9,316,887,841.6	9.176	0/1260
0.5	10	84,614,740,728.2	9,316,464,418.3	9.082	0/1260
0.5	100	83,669,473,895.1	9,316,006,663.5	8.981	0/1260
0.5	1000	83,473,127,668.6	9,315,911,580.9	8.960	0/1260
1.0	1	84,805,535,166.0	9,316,796,515.0	9.102	0/1260
1.0	10	84,274,682,479.6	9,316,445,647.7	9.046	0/1260
1.0	100	83,616,988,885.8	9,316,010,944.7	8.976	0/1260
1.0	1000	83,467,660,626.5	9,315,912,246.2	8.960	0/1260